

Multiple Choice

1. **Answer:** B

Options: A) Sender's Private key; B) Sender's Public key; C) Receiver's Private key; D) Receiver's Public key

Note: Correct option: B - Sender's Public key

2. **Answer:** C

Options: A) John the Ripper; B) L0phtCrack; C) Snort; D) Nessus

Note: Correct option: C - Snort

3. **Answer:** A

Options: A) Modification detection code (MDC); B) Modify authentication connection; C) Message authentication control; D) Message authentication cipher

Note: Correct option: A - Modification detection code (MDC)

4. **Answer:** B

Options: A) Assume that the query is satisfiable and continue executing the path.; B) Assume that the query is not satisfiable and stop executing the path; C) Restart STP and retry the query, up to a limited number of retries.; D) Remove a subset of the constraints and retry the query.

Note: Correct option: B - Assume that the query is not satisfiable and stop executing the path

5. **Answer:** D

Options: A) Yes, if the PRG is really "secure"; B) No, there are no ciphers with perfect secrecy; C) Yes, every cipher has perfect secrecy; D) No, since the key is shorter than the message

Note: Correct option: D - No, since the key is shorter than the message

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Authenticated'

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '56'

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Confidentiality, Integrity, Non repudiation and Authentication'.

Long Form Response

11. **Answer:** `def text_match_zero_one(text): | return 'Found a match!'`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def check_char(string): | return "Valid" | return "Invalid"`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key